BIBEK RAM

Chennai, India | +91 9939137481 | br4113@srmist.edu.in
LinkedIn | GitHub | ResearchGate

PROFESSIONAL SUMMARY

Biomedical Engineering graduate with strong exposure to medical device regulatory standards, quality systems, and manufacturing processes through experience at Bureau of Indian Standards (BIS). Skilled in medical device evaluation, technical documentation, and compliance with ISO 13485, FDA QSR, and CDSCO MDR. Combines engineering fundamentals with hands-on project experience in biomedical systems and edge-AI healthcare solutions. IEEE-published researcher with a strong interest in quality assurance, regulatory affairs, and medical device innovation..

EDUCATION

- **B.Tech – Biomedical Engineering (GPA: 9.75/10)**
  SRM Institute of Science and Technology, Chennai | 2022–2026
- **High School (+2 Science, GPA: 3.37/4.0)**
  St. Xavier's College, Kathmandu, Nepal | 2019–2021

TECHNICAL SKILLS

Programming: Python, MATLAB, C, TensorFlow, Scikit-learn, OpenCV
Biomedical Domains: Medical Imaging, EEG/ECG Analysis, Signal Processing, IoT Systems, Biosensors, Nanotechnology
Hardware & Systems: Arduino, Embedded Systems, PCB Design, Biomedical Instrumentation
Research Tools: PSD, Feature Extraction, Data Visualization, Statistical Analysis
Professional Skills: Leadership, Teamwork, Project Management, Critical Thinking

EXPERIENCE

Intern – Bureau of Indian Standards (BIS), New Delhi | June–July 2025- Certificate

- Contributed to pre-standardization of dialyzer reprocessing machines through technical evaluation, literature review, and field-level analysis
- Assisted in drafting Indian Standards (IS) covering performance, safety, and construction requirements aligned with regulatory frameworks
- Collaborated with industry experts from GK Healthcare, GrayFalcon, and Atlantic Biomedical , to understand compliance requirements, risk considerations, and quality benchmarks

Industrial Exposure – Panacea Medical Technologies Pvt. Ltd., Bangalore

- Learned about medical device manufacturing processes, quality control, and regulatory frameworks.

PROJECT HIGHLIGHTS

1. **Multimodal Tuberculosis Detection System (Ongoing)**
   AI-based TB screening using cough audio, chest X-ray, and clinical data with edge deployment on Raspberry Pi for early and predictive detection.
   Tech: Python, TensorFlow, Librosa, OpenCV, Raspberry Pi
2. **Edge AI–Based Multi-Model Healthcare Diagnostic Platform**
   Designed and deployed a unified Edge-AI system capable of running multiple healthcare diagnostic models locally on a Raspberry Pi for real-time medical analysis while ensuring low latency and data privacy.
   Tech: Python, TensorFlow Lite, TensorFlow, Flask, OpenCV, Raspberry Pi 4, Edge Computing
3. **Multimodal Emotion Recognition (EEG, Thermal & Facial Data)**
   Developed a multimodal emotion classification system combining EEG, thermal imaging, and facial data; achieved >95% accuracy for neuropsychiatric applications.
   Tech: Python, MATLAB, Machine Learning
4. **IoT Smart Helmet for Rider Safety GitHub Repo**
   Designed a smart helmet with crash detection, GPS tracking, vital sensing, air-quality monitoring, and cloud alerts using a custom PCB.
   Tech: Arduino, IoT, Sensors, PCB Design
5. **AI-Driven Closed-Loop Insulin Delivery System GitHub Repo**
   Built a closed-loop insulin pump prototype by modeling glucose–insulin dynamics and automating insulin delivery using control algorithms.
   Tech: MATLAB, Embedded Systems
6. **Smart Anti-Tremor Gloves GitHub Repo**
   Developed smart gloves with ML-based tremor detection and closed-loop vibration damping, enabling cloud-based remote neuro-rehabilitation.
   Tech: Arduino, Flex Sensors, IMU, Machine Learning, Cloud

7. **Temperature-Controlled Cooling System** **GitHub Repo**
   Implemented an automatic fan speed control system using real-time temperature sensing with live OLED display feedback.
   Tech: Arduino Uno, DS18B20, MOSFET

CERTIFICATIONS

- Master Medical Device Regulatory Affairs – Udemy (Jan 2026)- *Certificate*
- Mobile App Development With Kotlin[udemy 2025]- *Certificate*
- Machine Learning A–Z (Udemy, 2025)- *Certificate*
- Layer'24 Blockchain Hackathon – Blockchain Club SRM (Mar 2024)- *Certificate*
- Machine Learning Using Python – NIELIT Calicut (2024)- *Certificate*
- SRM Hackathon 8.0 – SRM Institute of Science and Technology | Apr 2024- *Certificate*
- AI in Healthcare – Great Learning (2023)- *Certificate*
- Fundamentals of Neuroscience for Neuroimaging – Johns Hopkins (2024)- *Certificate*
- Nanotechnology & Nanosensors – Technion (2023)- *Certificate*
- MATLAB Onramp – MathWorks (2023)- *Certificate*
- Python Programming – University of Michigan (2023)- *Certificate*
- Introduction to Healthcare – Stanford University (2024)- *Certificate*

PUBLICATION

"Multimodal Emotion Recognition via Correlation of EEG, Thermal, and Digital Image Data"
IEEE International Conference on Communication and Signal Processing (ICCSP 2025)
DOI: 10.1109/ICCSP64183.2025.11088814

LEADERSHIP & ACHIEVEMENTS

- **Student Ambassador, TekMedica Club (SRMIST):** Mentored peers in biomedical device design & innovation.
- **Healthathon 2024:** Presented *Smart Glass for Stress Detection*.
- **Workshop:** *Design Thinking for Sustainable Development* (2024).
- **Presenter:** *IEEE ICCSP 2025* (Paper on Multimodal Emotion Recognition).

LANGUAGES & INTERESTS

Languages: English, Hindi
Interests: Neuroengineering, Medical AI, Teaching, Innovation, Music